

COMP 3311

DATABASE MANAGEMENT

SYSTEMS

LECTURE 5 EXERCISES

RELATIONAL DATABASE DESIGN:

FUNCTIONAL DEPENDENCIES

EXERCISE 1

Assume that this relation instance contains the *only* set of tuples that may appear in $R(X, Y, V, W)$. Which of the following FDs hold in R ?

tuple	X	Y	V	W
1	x_1	y_1	v_1	w_1
2	x_1	y_1	v_2	w_2
3	x_2	y_1	v_1	w_3
4	x_2	y_1	v_3	w_4

- $X \rightarrow X$ **Yes** Trivial, holds in any relation.
- $X \rightarrow Y$ **Yes** For a given X value all Y values are identical.
- $X \rightarrow V$ **No** V values differ for same X value (e.g., tuples 1 & 2).
- $X \rightarrow W$ **No** W values differ for same X value (e.g., tuples 1 & 2).
- $Y \rightarrow X$ **No** X values differ for same Y value (e.g., tuples 2 & 3).
- $W \rightarrow X$ **Yes** All W values are different.
- $XV \rightarrow Y$ **Yes** X alone determines Y.
- $YV \rightarrow X$ **No** X values differ for same YV value in tuples 1 and 3.

EXERCISE 2

In Exercise 1, we assumed that we know all possible tuples in the relation instance, which is not usually true. In general, by looking at a relation instance, we can only tell FDs that are not satisfied.

List 5 FDs that are not satisfied in the relation instance.

A	B	C
a ₁	b ₁	c ₁
a ₁	b ₁	c ₂
a ₂	b ₁	c ₁
a ₂	b ₁	c ₃

$A \rightarrow C$

$B \rightarrow A$

$B \rightarrow C$

$C \rightarrow A$

$AB \rightarrow C$

EXERCISE 3

Given relation schema $R(X, Y, U, V, W)$ and $F = \{X \rightarrow Y, UV \rightarrow W, V \rightarrow X\}$

a) Determine the closure of each attribute.

$$X^+ = \{X, Y\} \quad (\text{Look for } X \text{ on LHS of FDs})$$

$$Y^+ = \{Y\}$$

$$U^+ = \{U\}$$

$$V^+ = \{V, X, Y\}$$

$$W^+ = \{W\}$$

b) What are the candidate keys of R ?

The candidate key is UV since $UV^+ = \{X, Y, U, V, W\}$.

EXERCISE 4

Given relation schema $R(A, B, C, D, E)$ and $F = \{A \rightarrow B, AB \rightarrow C, D \rightarrow AC\}$

a) Determine the following attribute closures.

$$A^+ = \{A, B, C\}$$

$$B^+ = \{B\}$$

$$C^+ = \{C\}$$

$$D^+ = \{D, A, B, C\}$$

$$E^+ = \{E\}$$

b) What are the candidate keys of R ?

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c) Find a canonical cover of F .

$$F_c = \{A \rightarrow B, AB \rightarrow C, D \rightarrow AC\} \quad \text{B extraneous in } AB \rightarrow C \text{ since } A \rightarrow AB \text{ \& } AB \rightarrow C \Rightarrow A \rightarrow C$$

$$F_c = \{A \rightarrow B, A \rightarrow C, D \rightarrow AC\} \quad \text{use union rule}$$

$$F_c = \{A \rightarrow BC, D \rightarrow AC\} \quad \text{C extraneous in } D \rightarrow AC \text{ since } D \rightarrow A \text{ \& } A \rightarrow C \Rightarrow D \rightarrow C$$

$$F_c = \{A \rightarrow BC, D \rightarrow A\}$$

EXERCISE 5

Given relation schema $R(A, B, C, G, H, I)$ and

$$F = \{A \rightarrow B, \quad A \rightarrow C, \quad CG \rightarrow H, \quad CG \rightarrow I, \quad B \rightarrow H\}$$

a) Is AG a (super)key of R given F? **Yes** **Why?** Since $AG \rightarrow R$

Compute AG^+

$$AG^{(0)} = \{A, G\}$$

$$AG^{(1)} = \{A, G, B\}$$

$$AG^{(2)} = \{A, G, B, C\}$$

$$AG^{(3)} = \{A, G, B, C, H\}$$

$$AG^{(4)} = \{A, G, B, C, H, I\}$$

$$(A \rightarrow B \text{ and } A \subseteq \{A, G\})$$

$$(A \rightarrow C \text{ and } A \subseteq \{A, G\})$$

$$(CG \rightarrow H \text{ and } CG \subseteq \{A, G, B, C\})$$

$$(CG \rightarrow I \text{ and } CG \subseteq \{A, G, B, C, H\})$$

b) Is AG a candidate key? **Yes** **Why?** Cannot remove A or G

c) Does $A^+ \rightarrow R$ hold? **No** since $A^+ = \{A, B, C, H\}$

d) Does $G^+ \rightarrow R$ hold? **No** since $G^+ = \{G\}$

EXERCISE 6

We want to create the database for a bank that contains accounts (A), branches (B) and customers (C).

a) What are the functional dependencies implied by the following constraints?

- i. An account cannot be shared by multiple customers.

Account $\rightarrow$ Customer $A \rightarrow C$

- ii. Two different branches do not have the same account.

Account $\rightarrow$ Branch $A \rightarrow B$

- iii. Each customer can have at most one account in a branch (but different accounts in different branches).

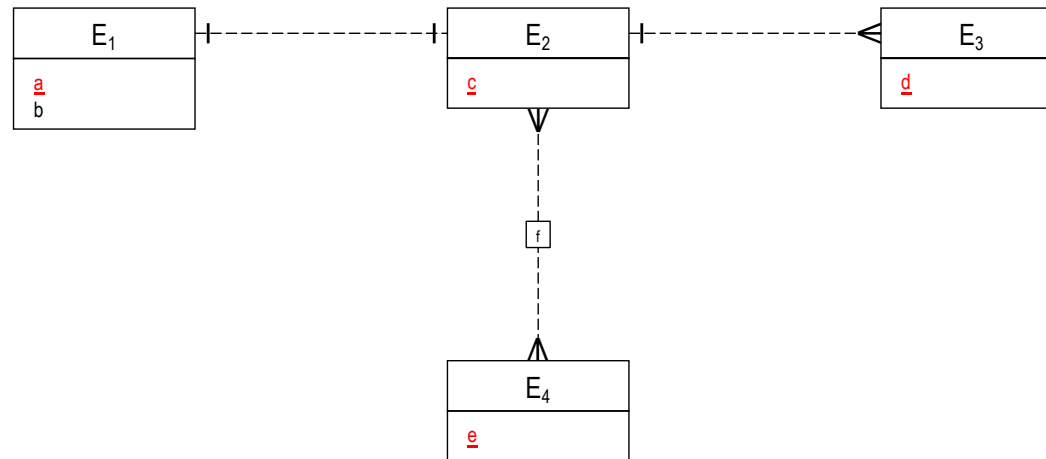
Branch, Customer $\rightarrow$ Account $BC \rightarrow A$

b) What are the candidate keys?

(Branch, Customer) and Account BC and A

EXERCISE 7

What are the FDs implied by the E-R diagram if all relationship constraints are maximum cardinalities?



FDs implied by the E-R diagram:

$a \rightarrow b$
 $a \rightarrow c$
 $c \rightarrow a$
 $d \rightarrow c$
 $ce \rightarrow f$

FDs obtained using inference rules:

$c \rightarrow b$ (from $c \rightarrow a$ and $a \rightarrow b$) **IR3**
 $d \rightarrow a$ (from $d \rightarrow c$ and $c \rightarrow a$) **IR3**
 $d \rightarrow b$ (from $d \rightarrow c$ and $c \rightarrow b$) **IR3**